

HALEEMA TALLAT

Machine Learning Engineer · AI evaluations, agents, LLM inference

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AI SAFETY & EVALUATIONS

LeakBench — github.com/haleematalat/leakbench

2026

- Built a 30-task Inspect benchmark that tests whether AI agents can find and fix the bugs that make an ML evaluation report the wrong number, plus 5 correct pipelines that catch over-editing. Agents work in network-isolated Docker sandboxes and are scored by hidden checks.
- Made every task verifiable before any model ran: 120 no-model checks confirm each one is solvable and fails when the data is edited or the tests are faked. Found and fixed scoring ranges that rejected valid fixes or passed half-fixes.
- Showed it separates models: Qwen 3.5 9B solved 75% of bug tasks against 29% for the 4B (95% CI +29 to +61 points). Without a hint that the eval might be wrong, the 4B fell to 12%.

ConspiracyBench — Stationhouse, research collaborator · [Hugging Face](#)

2026 – Ongoing

- Found the benchmark's headline divergence rate was inflated from about 20% to 49%: Qwen 3.5 4B used up its token budget on reasoning and returned empty answers on half its turns, and the LLM judge still graded them.
- Fixed it in the scorer: empty answers are now marked invalid instead of judged, halving judge calls. Another contributor reproduced the issue on two more judge models.
- Built a log-analysis tool that shows where a run's time goes and checks byte for byte that speed changes leave prompts unchanged; it traced 51 timed-out requests per run to a concurrency default.

Coursework: Technical AI Safety, BlueDot Impact (2026) · ARENA curriculum, self-paced (2026)

EXPERIENCE

Senior AI Engineer, Agents & LLM Infrastructure — [IgniteTech \(via Crossover\)](#)

Jul 2025 – Present

- Designed and built IRIS's first production social engine in ~3 months, handling 1,000+ customer interactions a day across 10–15 channels: multi-tenant Node.js/TypeScript services for Instagram, Facebook and LinkedIn ingestion, case processing and orchestration on AWS. Its playbook and runbooks became the template for 20 more engines.
- Scaled the engines to 27 on Kafka with KEDA autoscaling and graceful draining, and cut memory 6–10× and startup ~20× by replacing Java monoliths with stateless TypeScript services.
- Architected a distributed platform running up to 500 concurrent coding agents: LangGraph on Ray, Pydantic-typed state and tools, Kafka events, Redis and PostgreSQL state, on Kubernetes on AWS.
- Made agent runs measurable by tracing every model and tool call in Langfuse, tracking cost per run and tool-call errors in ClickHouse, and scoring drafted replies by acceptance and edit rates.
- Built the LLM inference layer for agent runs and AI-drafted replies, pairing self-hosted vLLM and Hugging Face models with the Anthropic API; cut cost and latency with batching, quantisation and context management that makes agents reuse existing code.
- Designed the governed MCP architecture for Claude Code, Cursor and the agent platform, with OAuth2, capability controls, output validation and human approval for high-impact actions; shipped on it an agent that checks live requests against schema-encoded rules.

Computer Vision Engineer (Contract) — [Brick & Mortar AI](#)

Mar – Oct 2025

- Showed a few-shot classifier reported at 46.7% actually scored 89.7% by tracing split leakage and protocol mismatches in its eval harness and rebuilding the held-out split.
- Cut cross-camera identity switches 20% and raised throughput 10–15× with segmentation-masked ReID (YOLOv8/v11) served on Triton with TensorRT.

Computer Vision Engineer — [NextBridge](#)

Nov 2024 – Mar 2025

- Built a smart wheelchair's end-to-end WebRTC control stack: signalling server (SDP/ICE), joystick sender and on-chair Jetson ROS 2 receiver.
- Made the link fail-safe with heartbeats, backoff reconnection and a watchdog that stops the chair on link loss; instrumented round-trip latency.

Machine Learning Engineer — [Powersoft19](#)

Nov 2023 – Nov 2024

- Cut field engineers' problem-resolution time 90% with a production hybrid RAG system over technical documents, measured with recall@k.
- Cut fault-detection turnaround from ~48 hours to ~30 minutes by leading 3 engineers to replace manual inspection with anomaly detection on 1TB+ of device telemetry.

Research Engineer — [KICS, Center for Language Engineering, UET](#)

Nov 2022 – Nov 2023

- Reached 93% accuracy on NLP classification of enterprise text, and reproduced RAG methods from the literature.
- Built Tajziyakar, a containerised OCR pipeline that extracts and flags TV news tickers at archive scale.

EDUCATION & AWARDS

BSc Electrical Engineering, UET Lahore (2023) · CureMD Master Inventor Award (2022) · COHACK winner (2022)

SKILLS

Evaluations: Inspect, benchmark and task design, Docker sandboxes, LLM judges, paired bootstrap, Langfuse

LLMs & agents: LangGraph, Ray, MCP, vLLM, Hugging Face, Anthropic API, RAG, Qdrant, fine-tuning

Infrastructure: AWS, Kubernetes, Docker, Kafka, KEDA, Redis, PostgreSQL, ClickHouse, Elasticsearch, Airflow

Languages & ML: Python, TypeScript, Node.js, React, SQL, C++, PyTorch, CUDA, ONNX, TensorRT, Triton